



Problem solving with non-linear relationships

1 Which of these are triangular numbers? Tick **2** correct answers

☐ 0

☒ 1

☐ 5

☒ 10

☐ 20

2 Use a calculator to work out the value of 3^8 . 6561. Fill in the blank

3 The next term in the geometric sequence 0.4, 2, 10, ... is 50. Fill in the blank

4 Which of these could be the first 4 terms in an arithmetic sequence? Tick **2** correct answers

☐ 28, 22, 16, 11, ...

☐ 136, 163, 190, 209, ...

☒ 13, 49, 85, 121, ...

☒ 17, 9, 1, -7, ...

☐ -21, -8, 8, 21, ...

5 What is the n^{th} term rule for the linear sequence which starts 13, 8, 3, -2, ...? Tick **1** correct answer

☒ $18 - 5n$

☐ $13 - 5n$

☐ $8 - 5n$

☐ $5n + 8$

☐ $5n + 13$



6 Which of these could be the first 5 terms of a sequence with a common second difference? (These are called quadratic sequences). Tick **3** correct answers



2, 3, 5, 8, 12, ...



8, 12, 18, 28, 40, ...



9, 10, 12, 16, 24, ...



12, 17, 27, 42, 62, ...



15, 21, 28, 36, 45, ...

